NAME ______________________ DATE ______________________ PERIOD ______________________

SOLAR AGRICULTURAL AGENCY · STUDENT MISSION

CASE 01 · ISS GREENHOUSE MODULE

Campaign 1 · Low Earth Orbit · Gravitropism in Microgravity

VALIDATION BUILD

MISSION QUESTION

Why are the lettuce roots tangled and directionless even though water, nutrients, temperature, and lighting appear normal?

SCIENCE FOCUS · GRAVITROPISM IN MICROGRAVITY

Plants normally use gravity to help guide root and shoot growth. In microgravity, that directional cue becomes unreliable, so engineers must provide another way for plants to orient.

Science status: The ISS case story is fictional. Gravitropism, statoliths, and plant responses to environmental cues are real.

REFERENCE

1 · Vocabulary

Gravitropism	Directional growth in response to gravity.	**Microgravity**	Apparent weightlessness during continuous free fall in orbit.
Statocyte	Plant cell involved in sensing gravity.	**Statolith**	Dense starch-filled structure that helps provide a direction cue.
Phototropism	Directional growth in response to light.		

PREDICTION

2 · Initial thinking

What evidence would help you distinguish a resource problem from an orientation problem?

Name the reading, observation, or record that would help.

Evidence, Causes & Mechanism

EVIDENCE TASK

3 · Investigate four evidence sources

Record only evidence that helps support or reject a cause. O = observation/data; I = inference/explanation.

Source	Observation or data	What does it support or reject?	O / I
Crew			
Sensors			
Plants			
Mission logs			

CAUSE TEST

4 · Test the competing explanations

Explanation	Status	One evidence-based reason
Nutrient solution is damaging roots.		
The light cycle is wrong.		
The seed stock is defective.		
Microgravity disrupts orientation.		

MECHANISM MODEL

5 · Build the mechanism

WORD BANK curve or grow without consistent orientation · downward · settle · settle in one direction

Earth gravity

Statoliths ______

→ consistent direction signal

→ roots grow ______

Microgravity

Statoliths do not ______ consistently

→ unreliable direction signal

→ roots ______

Diagnosis & Design Response

DIAGNOSIS

6 · Diagnose and reject an alternative

State the cause in your own words.

Use the environmental condition and the plant mechanism.

Reject the most tempting alternative using evidence.

Name the alternative and the evidence that weakens it.

EXPLANATION

7 · Claim-Evidence-Reasoning

CLAIM	
EVIDENCE	
REASONING	

ENGINEERING RESPONSE

8 · Supply a consistent orientation cue

Choose or design root channels/mesh, a moisture gradient, shoot lighting plus a root guide, a rotating chamber, or another justified design.

Design	One criterion	One constraint

TRANSFER CHECK

9 · Exit ticket

Engineers add narrow root channels but do not change the lighting. Which symptom should improve first: tangled roots or pale leaves? Explain.

OPTIONAL EXTENSION

Compare a physical root guide with a rotating growth chamber. Which is easier to use on the ISS, and which more closely replaces gravity?

Student source line: NASA Plant Biology and Plant Gravity Perception resources; NGSS MS-ETS1 Engineering Design.

ISS GREENHOUSE MODULE

Campaign 1 · Case 1 · Gravitropism in Microgravity · 60 minutes

VALIDATION BUILD

PREPARATION

Before class

- Confirm the game loads.
- Print the student sheet double-sided or use fillable HTML mode.
- Keep the evidence fallback ready.
- Provide headphones.

GAME ACCESS

Launch path

interstellarharvest.github.io/Space-Sprout-Sleuth/

Select Campaign 1 → Case 1.

CORRECT DIAGNOSIS

Central interpretation

Microgravity disrupts gravitropism. Roots need an alternative consistent cue or guidance system.

LESSON FLOW

60-minute plan

Time	Action
0–5	Launch game and materials
5–10	Initial prediction
10–30	**3 · Investigate four evidence sources**
30–38	**4 · Test the competing explanations**
38–48	**6 · Diagnose and reject an alternative + 7 · Claim-Evidence-Reasoning**
48–55	**8 · Supply a consistent orientation cue**
55–60	**9 · Exit ticket**

ESSENTIAL EVIDENCE

Evidence students should notice

Source	Essential evidence
Crew	Roots do not grow down; resource adjustments did not solve the pattern.
Sensors	Microgravity is the major unusual condition; nutrients, water, temperature, humidity, and lighting are nominal.
Plants	Dense tangles, poor anchoring, and unusual stem angles.
Logs	Gravitropism/statolith mechanism and possible replacement cues.

LIKELY STICKING POINT

Microgravity is not the literal absence of gravity. The ISS and everything inside it are continuously falling around Earth, producing apparent weightlessness.

COLLECTION

What to collect

3 · Investigate four evidence sources · 6 · Diagnose and reject an alternative · 7 · Claim-Evidence-Reasoning · 8 · Supply a consistent orientation cue · 9 · Exit ticket

FALLBACK

Technical fallback

Use the evidence digest on the final teacher page. Students complete the same academic task without the game.

TEACHER LINE

“Your job is not to click everything. Your job is to determine which evidence best explains the root pattern.”

ISS Greenhouse Module

OVERVIEW

Lesson overview

Students investigate a stunted lettuce crop aboard the International Space Station. They compare crew testimony, environmental sensors, direct plant observations, and mission logs to determine why roots form tangled balls despite normal resources. Students explain the gravitropism mechanism and evaluate an engineered alternative cue.

> GUIDING QUESTION
> Why are the lettuce roots tangled and directionless even though water, nutrients, temperature, and lighting appear normal?

ALIGNMENT

Standards alignment

- **MS-ETS1-1:** Supports defining criteria and constraints for a root-guidance solution.
- **MS-ETS1-2:** Supports evaluating competing orientation solutions.
- **SEP:** Constructing Explanations and Designing Solutions; Analyzing and Interpreting Data.
- **DCI:** ETS1.A and ETS1.B, with plant structure/function as supporting science.
- **CCC:** Cause and Effect; Structure and Function.

This lesson supports components of the standards; it does not by itself fulfill an entire performance expectation.

OUTCOMES

Learning objectives

1. Distinguish observations from explanations across four evidence sources.
2. Identify disrupted gravitropism as the best-supported cause.
3. Explain statolith movement as part of a directional gravity-sensing mechanism.
4. Reject an alternative explanation using evidence.
5. Define a criterion and constraint for an orientation system.

SUCCESS CRITERIA

What proficient work includes

Relevant evidence from at least three sources · correct mechanism · evidence-based alternative rejection · concise CER · directional engineering response.

REFERENCE

Vocabulary

gravitropism · microgravity · statocyte · statolith · phototropism

PREPARATION

Materials

Chromebooks · headphones · game access · double-sided mission sheet or editable HTML · evidence fallback.

NOTES

Procedure

PREPARATION

Teacher preparation

1. Open the game and confirm Campaign 1, Case 1 loads.
2. Test audio/headphones.
3. Print or share the student sheet.
4. Keep the evidence fallback ready.

PROCEDURE

0–5 minutes · Setup

Students launch the game and open the mission sheet. State that game score, rank, speed, and optional dialogue are not academically graded.

PROCEDURE

5–10 minutes · Phenomenon launch

Show the greenhouse image or read the briefing. Ask what evidence would distinguish a resource problem from an orientation problem. Students complete **2 · Initial thinking**; do not confirm the diagnosis.

PROCEDURE

10–30 minutes · Gameplay investigation

Students visit crew, sensors, plants, and logs. They record only evidence that supports or rejects a cause.

> FACILITATION PROMPTS
> Which detail is a symptom? Which normal reading rules out an explanation? Have you used more than one source?

PROCEDURE

30–38 minutes · Evidence checkpoint

Students complete **4 · Test the competing explanations**. Confirm that resource readings are normal and that the root pattern is directional and repeated. Do not require optional dialogue completion.

PROCEDURE

38–48 minutes · Diagnosis and CER

Students complete **6 · Diagnose and reject an alternative** after the game unlocks, then complete **7 · Claim-Evidence-Reasoning** using evidence from several sources and reasoning that links microgravity to a weakened directional cue.

PROCEDURE

48–55 minutes · Engineering application

Students choose an alternative cue and give one criterion and constraint. Strong responses distinguish a direct root cue from lighting that primarily guides shoots.

PROCEDURE

55–60 minutes · 9 · Exit ticket

Students answer independently. Collect or save the mission file.

Assessment, Access & Accuracy

FORMATIVE CHECKS

Checks for understanding

- Separates symptom from cause.
- Uses normal readings to reject alternatives.
- Describes microgravity as apparent weightlessness.
- Links statolith direction to root orientation.
- Proposes a cue that addresses the problem.

ASSESSMENT

Evidence of learning

Formative: prediction, evidence matrix, explanation ranking, teacher questions.

Collected: diagnosis, rejected alternative, CER, engineering response, exit ticket.

ACCESS

Accessibility supports

- Bullet-point evidence accepted.
- CER frame available but optional.
- Text-to-speech compatible.
- Game evidence may remain open.
- Evidence fallback available.
- No group product required.

MISCONCEPTIONS

Common misconceptions

1. **No gravity:** The ISS remains under gravity while continuously falling around Earth.
2. **Roots use only gravity:** Roots also respond to moisture, touch, chemicals, and other cues.
3. **Brighter light fixes roots:** Light is a stronger shoot cue; roots need a more direct guide.
4. **Plants cannot grow in space:** Growth is possible with adapted environmental systems.
5. **Plant pillows recreate gravity:** They manage media and fluids, not a gravity vector.

SOURCE STATUS

Science status

ESTABLISHED
gravitropism, statolith involvement, phototropism, plant responses to microgravity.

PLAUSIBLE ENGINEERING
root channels, moisture cues, rotating systems. Students complete **8 · Supply a consistent orientation cue** on the worksheet; Campaign 1 has no in-game apply-the-fix step, so these solutions appear only as discussion in the mission logs, not as a selectable action.

FICTIONAL CONTEXT
the specific emergency, crew, SAA, and exact symptoms.

FALLBACK

Technical fallback

Distribute the evidence digest. Students complete the same mission sheet; technical failure is not graded.

NOTES

Evidence Architecture & Distractors

EVIDENCE ARCHITECTURE

Evidence map

Source	Essential evidence	Role
Crew	Roots do not grow down; changing resources did not help.	Symptom and failed interventions
Sensors	Microgravity; normal nutrients, water, climate, and lighting.	Strong causal difference and elimination evidence
Plants	Tangled roots, poor anchoring, unusual stems.	Direct biological observation
Logs	Gravitropism/statolith mechanism and possible replacement cues.	Mechanism and design options

REASONING PATH

How the case resolves

1. Root tangling is the symptom.
2. Normal resources weaken simple resource explanations.
3. Microgravity is the major unusual environmental condition.
4. Logs connect gravity sensing to directional growth.
5. Disrupted gravitropism best explains the pattern.
6. A useful solution supplies another consistent cue.

DISTRACTORS

Competing explanations

Distractor	Why tempting	Why rejected
Nutrient burn	Pale, stunted plants	Nominal solution; directional tangling is a poor match
Wrong light cycle	Odd stems and pale leaves	16/8 and PAR are nominal; roots are the strongest abnormality
Defective seeds	All plants affected	Pattern repeats with environment and prior plantings
Water problem	Roots seek water	Delivery works; uniform wicking may reduce a cue but does not best explain the case

The nutrient, light, and seed distractors are the three incorrect options students can select on the game's Diagnose screen. "Water problem" is an additional teacher discussion distractor for class conversation — it is not one of the selectable in-game diagnoses.

BOUNDARY NOTE

The game simplifies a complex plant response. Do not teach that roots are completely random, that statoliths are the only gravity-sensing mechanism, or that plant pillows fully solve orientation.

INSTRUCTIONAL IDENTITY

Student role and primary practice

Student role: Pattern investigator. **Primary practice:** compare multiple evidence types, eliminate alternatives, explain a mechanism, and define an engineering problem.

NOTES

Quick & Analytic Rubrics

CHECKLIST RUBRIC

Quick classroom rubric

Dimension	Secure	Developing	Beginning
Evidence	Specific, relevant evidence from several sources	Some relevant evidence; important detail/source missing	Vague, copied, or irrelevant evidence
Mechanism	Connects microgravity, statolith direction, and root orientation	Names gravitropism but mechanism is incomplete	Cause stated without mechanism
Reasoning	Connects evidence to claim and rejects an alternative	Partial connection or weak rejection	Evidence listed without explanation
Communication	Accurate vocabulary and complete product	Minor vocabulary/completion gaps	Major gaps impede meaning

ANALYTIC GRID

Formal analytic rubric

Criterion	4 · Accomplished	3 · Proficient	2 · Developing	1 · Beginning
Evidence selection	Most relevant; symptom/cause distinguished	Relevant with minor omissions	Mixed relevant/irrelevant	Little usable evidence
Source diversity	Integrates 3–4 source types	Uses at least 2 source types	Mostly one source	Sources unclear
Mechanism	Accurate and appropriately qualified	Central mechanism accurate	Important gaps	Unsupported/inaccurate
Alternative evaluation	Specific contradictory evidence	Generally relevant evidence	Weak support	No evaluation
Engineering	Feasible cue, criterion, constraint	Relevant solution + criterion/constraint	Loosely related solution	Does not address orientation
Communication	Clear, concise, complete	Mostly clear/complete	Inconsistent but understandable	Incomplete/difficult

SCORING BOUNDARY

No numeric points appear on the student page by default. Game score, rank, speed, and optional exploration are never academic criteria.

NOTES

ISS Greenhouse Module

AUTHORITATIVE SOURCES

Source list

1. NASA, *Growing Plants in Space* nasa.gov/exploration-research-and-technology/growing-plants-in-space/
2. NASA Ames, *Plant Gravity Perception (Space-13)* nasa.gov/ames/space-biosciences/plant-gravity-perception-spacex-13/
3. NASA, *Veggie Plant Growth System Activated on ISS* nasa.gov/missions/station/veggie-plant-growth-system-activated-on-international-space-station/
4. NASA Science, *Plant Biology Program · Hardware* science.nasa.gov/biological-physical/focus-areas/plant-biology/hardware/
5. Ditengou, Teale & Palme (2020), *Settling for Less: Do Statoliths Modulate Gravity Perception?* pubmed.ncbi.nlm.nih.gov/31963631/
6. NASA OSDR, Arabidopsis Seedlings in Microgravity (OSD-38) osdr.nasa.gov/bio/repo/data/studies/OSD-38
7. NGSS MS-ETS1-1 and MS-ETS1-2 nextgenscience.org

EVIDENCE FALLBACK

No-game evidence digest

Briefing: ISS lettuce is stunted; roots curl around plant pillows instead of anchoring.

Crew: roots do not grow downward; changing resources did not help.

Sensors: microgravity; nutrients, water, temperature, humidity, and lighting are nominal.

Plants: dense tangles, poor anchoring, and angled stems.

Logs: statolith-based gravity sensing normally provides direction; possible compensations include root channels, moisture gradients, shoot lighting, or rotating systems.

TECHNICAL NOTES

Classroom fallback

- Refresh once if the game does not load.
- Use a current Chromebook Chrome tab.
- The Diagnose screen unlocks after formal clues are collected.
- Do not require every optional conversation branch.
- Move to the fallback within two minutes rather than losing lesson time.

NOTES

EVIDENCE & ALTERNATIVES

Case 01 · ISS Greenhouse Module · Teacher-only model responses

VALIDATION BUILD

EVIDENCE TASK

3 · Investigate four evidence sources

Source	Model evidence	Interpretation
Crew	Roots grow in many directions; nutrient, water, and light adjustments did not fix them.	Supports orientation problem; weakens simple resource causes.
Sensors	Microgravity; other major readings nominal.	Identifies unusual condition and rejects several alternatives.
Plants	Dense tangles, wrapping, weak anchoring, odd stems.	Direct symptom of disrupted directional growth.
Logs	Statolith/gravitropism mechanism and replacement cues.	Explains cause and provides engineering options.

CAUSE TEST

4 · Test the competing explanations

- **Nutrient damage · weakened:** readings are nominal; tangling is not a strong match.
- **Wrong light cycle · weakened:** schedule/intensity are nominal; lighting does not best explain roots.
- **Defective seeds · weakened:** repeated environment-linked pattern.
- **Microgravity · supported:** matches all four evidence sources and the mechanism.

NOTES

Mechanism & Diagnosis

MECHANISM MODEL

5 · Build the mechanism

Earth gravity → statoliths **settle** → consistent signal → roots grow **downward**.

Microgravity → statoliths do not **settle in one direction** → unreliable signal → roots **curve or grow without consistent orientation**.

ACCURACY NOTE

Avoid requiring “random.” Roots may still respond to moisture, light, touch, chemicals, and internal growth programs.

DIAGNOSIS

6 · Diagnose and reject an alternative

Microgravity disrupted normal gravitropic orientation. Without a stable settling direction for statoliths, roots lacked a reliable “down” cue and formed tangled growth around the plant pillows.

REJECTED ALTERNATIVE

6 · Diagnose and reject an alternative

Nutrient damage is tempting because the plants are stunted and pale, but the sensors show nominal nutrient levels. It also does not explain the roots’ consistent directional/tangling problem.

NOTES

CER, Design & Exit Ticket

CLAIM-EVIDENCE-REASONING

7 · Claim-Evidence-Reasoning

CLAIM	The crop problem was caused by disrupted gravitropism in microgravity.
EVIDENCE	Kim reported roots that did not grow downward. Sensors showed microgravity while resources were normal. Plant inspection showed dense tangles and poor anchoring. Logs connected statolith-based sensing to root direction.
REASONING	Gravitropism depends on a consistent gravity direction. In microgravity, statoliths do not settle in one stable direction, so roots lose a reliable orientation signal. This explains the root pattern better than nutrient, lighting, or seed explanations, which are weakened by normal readings and the repeated environmental pattern.

ENGINEERING RESPONSE

8 · Supply a consistent orientation cue

Example design: Directional root channels inside the plant pillow. **Criterion:** most roots should extend into the medium rather than wrapping around the surface. **Constraint:** channels must remain lightweight and permit water and oxygen flow.

Directional light alone is a weaker root solution but can support shoot orientation as part of a combined design.

EXIT TICKET

9 · Exit ticket

Expected: Tangled roots should improve first because channels directly guide root direction and anchoring. Pale leaves may improve later if root structure improves resource access.

ACCEPTABLE VARIATION

Scoring guidance

- Students may use “apparent weightlessness,” “microgravity,” or “reduced reliable gravity cue.”
- Students do not need to describe auxin unless taught separately.
- A rotating chamber, moisture gradient, or combined guide can earn full credit when criteria and constraints are defensible.

NAME ______ DATE ______ PERIOD ______

SOLAR AGRICULTURAL AGENCY · ACCESSIBLE STUDENT MISSION

CASE 01 · ISS GREENHOUSE MODULE

Parallel accessible edition · Same investigation and evidence sequence

VALIDATION BUILD

MISSION QUESTION

Why are the lettuce roots tangled and directionless even though water, nutrients, temperature, and lighting appear normal?

SCIENCE FOCUS

Plants normally use gravity to help guide root and shoot growth. In microgravity, that directional cue becomes unreliable, so engineers must provide another way for plants to orient.

REFERENCE

1 · Vocabulary

Gravitropism	Directional growth in response to gravity.
Microgravity	Apparent weightlessness during continuous free fall in orbit.
Statocyte	Plant cell involved in sensing gravity.
Statolith	Dense starch-filled structure that helps provide a direction cue.
Phototropism	Directional growth in response to light.

PREDICTION

2 · Initial thinking

What evidence would help you tell a resource problem from an orientation problem?

You may answer with words, phrases, or bullet points.

Evidence Sources

EVIDENCE TASK

3 · Investigate four evidence sources

Record only evidence that helps support or reject a cause. You may use phrases or bullet points.

Source	Observation or data	What does it support or reject?
Crew		
Sensors		
Plants		
Mission logs		

NOTES

Competing Explanations & Mechanism

CAUSE TEST

4 · Test the competing explanations

For each explanation, mark supported or weakened. Then give one reason.

Nutrient solution is damaging roots.

Status and one evidence-based reason.

The light cycle is wrong.

Status and one evidence-based reason.

The seed stock is defective.

Status and one evidence-based reason.

Microgravity disrupts orientation.

Status and one evidence-based reason.

MECHANISM MODEL

5 · Build the mechanism

WORD BANK curve or grow without consistent orientation · downward · settle · settle in one direction

Earth gravity	**Microgravity**
Statoliths ______	Statoliths do not ______ consistently.
Roots grow ______	Roots ______

Diagnosis & Alternative

DIAGNOSIS

6 · Diagnose and reject an alternative

What caused the tangled, directionless roots?

Use the environmental condition and the plant mechanism.

ALTERNATIVE

6 · Diagnose and reject an alternative

Which other explanation is tempting, and what evidence weakens it?

You may answer in two sentences or bullet points.

NOTES

Claim-Evidence-Reasoning

EXPLANATION

7 · Claim-Evidence-Reasoning

You may write sentences or use bullet points. Use evidence from more than one source.

CLAIM	
EVIDENCE	
REASONING	

Engineering Response & Exit Ticket

ENGINEERING RESPONSE

8 · Supply a consistent orientation cue

Choose or design a system that gives roots a consistent cue.

Describe or draw your design.

Examples include root channels, a moisture gradient, a root guide plus lighting, or a rotating chamber.

One criterion

What must the design accomplish?

One constraint

What limit must the design work within?

TRANSFER CHECK

9 · Exit ticket

Engineers add narrow root channels but do not change the lighting. Which symptom should improve first: tangled roots or pale leaves? Explain.

OPTIONAL EXTENSION

Compare a physical root guide with a rotating growth chamber. Which is easier to use on the ISS, and which more closely replaces gravity?